

SOCrates

- **Problem Statement ID – 26157**
- **Problem Statement Title – Supervisory**

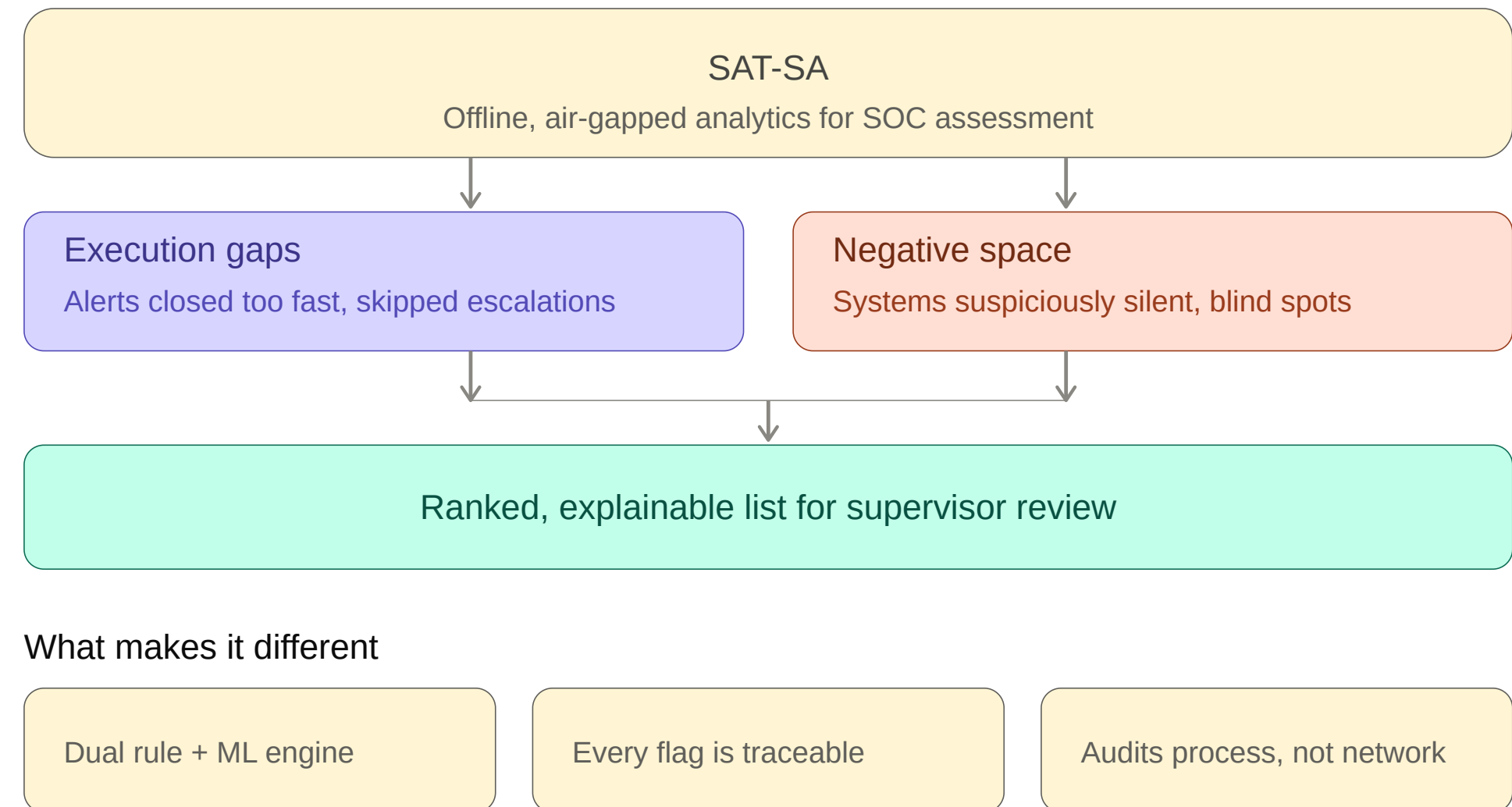
Analytics Tool for SOC Assessment (SAT-SA)

- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID –**
- **Team Name – SOCrates**



* Proposed Solution

- SAT-SA runs offline and reads SOC alert metadata without touching a network.
- Two detection engines work together: Fixed rules catch known patterns, and an Isolation Forest model catches the anomalies rules might miss.
- Picks up execution gaps: alerts closed too quickly, escalations skipped, copy-pasted investigation notes.
- Checks for negative gaps: a critical system that hasn't generated a single alert is the biggest red flag.
- Every flag marked has evidence with it, so a supervisor can see exactly why.
- Most SOC research focuses on catching threats faster; we're auditing whether the SOC's own review process is actually doing its job.

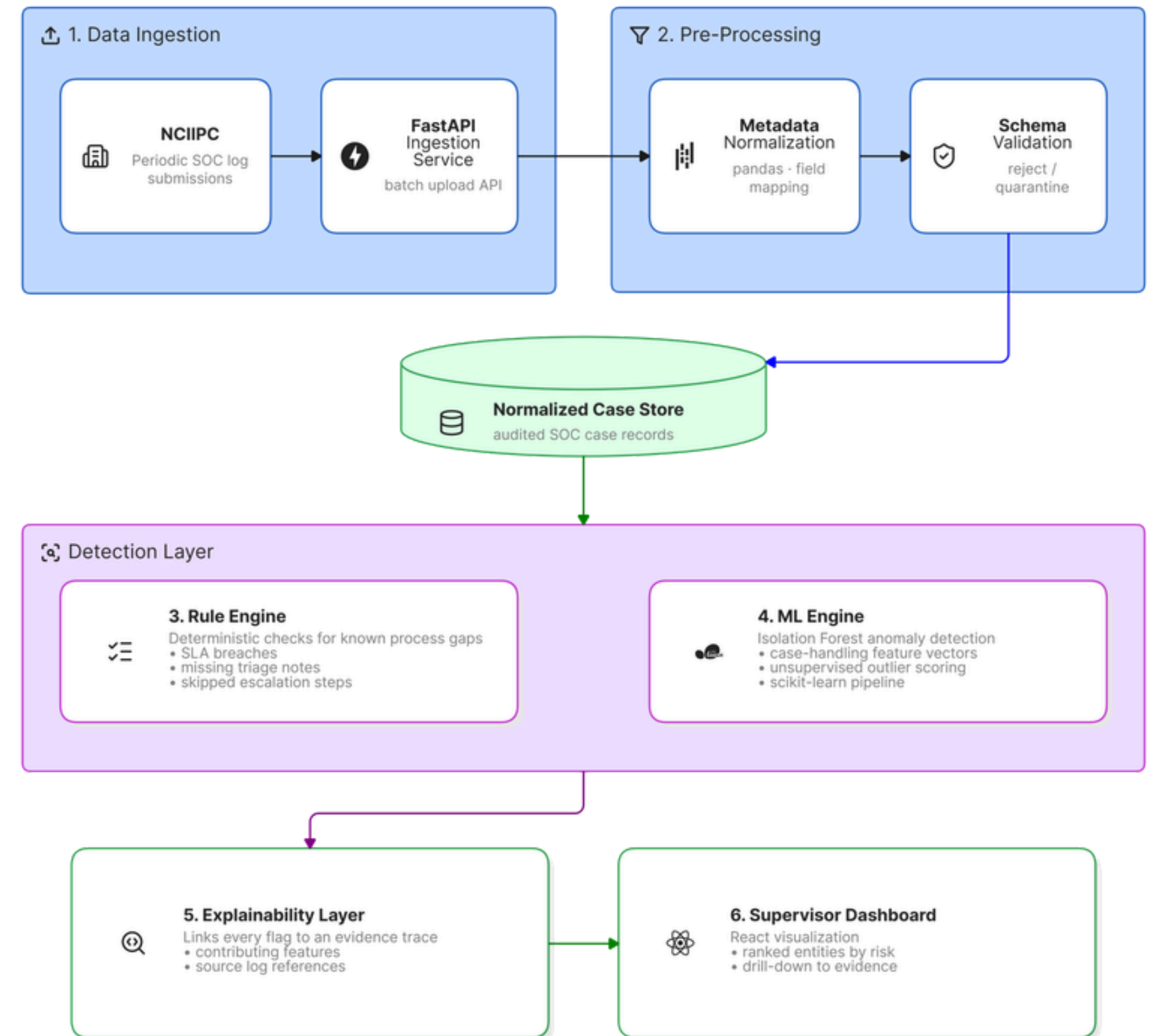
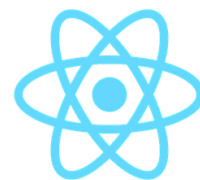


Tech Stack:

- Python
- FastAPI
- React
- scikit-learn (Isolation Forest)
- pandas

Methodology:

1. Ingest SOC alert/case data
2. Normalize into a unified schema
3. Rule-based engine flags execution gaps & negative space patterns
4. Isolation Forest flags statistical anomalies that rules might miss
5. Explainability layer attaches evidence-based reasoning to every flag
6. Dashboard presents ranked, prioritized findings to supervisors



» Metadata-Only Processing:

- Analyzes structured alert and ticket logs (timestamps, notes, severity) rather than raw packet captures.

» Native Fit for NCIIPC:

- Matches the existing periodic, batch model used by Critical Sector Entities.

» Viability:

- **Zero Infrastructure Overhead:** Deploys as a self-contained offline application. No cloud dependencies required.
- **National Scalability:** Enables a small supervisory team to audit dozens of entities simultaneously.
- **Assists Human Judgment:** Designed as an investigative aid to highlight high-risk discrepancies.

» Key Challenges & Risks:

- **Threshold Calibration:** Balancing sensitivity across entities of drastically different sizes and alert volumes.
- **Data Discrepancies:** Transitioning from synthetic data to heterogeneous real-world entity formats.
- **Risk of False Positives:** Flagging valid operational anomalies (e.g. scheduled maintenance) as suspicious.

Metadata-Only Processing

Analyzes structured alert and ticket logs — timestamps, notes, severity — rather than raw packet captures.

Native Fit for NCIIPC

Matches the existing periodic, batch model already used by Critical Sector Entities for supervisory submissions.

Viability

1 Zero Infrastructure Overhead

Deploys as a self-contained, offline application — no cloud dependencies required.

2 National Scalability

Enables a small supervisory team to audit dozens of entities simultaneously.

3 Assists Human Judgment

Designed as an investigative aid that highlights high-risk discrepancies for review.

Key Challenges & Risks

+ Threshold Calibration

Balancing sensitivity across entities of drastically different sizes and alert volumes.

+ Data Discrepancies

Transitioning from synthetic data to heterogeneous real-world entity formats.

+ Risk of False Positives

Flagging valid operational anomalies — e.g. scheduled maintenance — as suspicious.



IMPACT AND BENEFITS



1. NCIIPC Supervisors:

- Scalable Inspection: Audits 100% of submitted metadata without increasing manual review hours.
- Objective Prioritization: Automatically surfaces high-risk entities through aggregate risk score rankings.
- Defensible Action: Built-in explainability gives supervisors direct log evidence for every flag.

2. Critical Sector Entities:

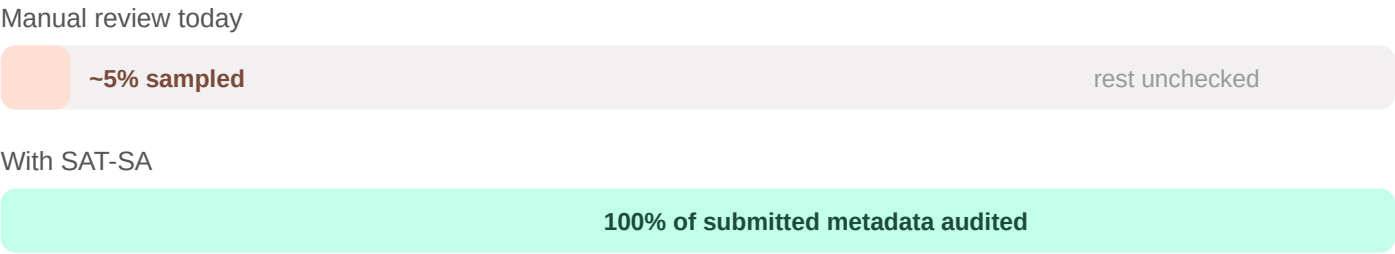
- Drives True Security: Identifies superficial compliance and incentivizes genuine threat triage over speed.
- Uncovers Blind Spots: Exposes internal SOC bottlenecks, skipped escalations, and unmonitored assets.
- Consistent Standards: Provides uniform, objective evaluation benchmarks across all critical sectors.

3. National Cyber Governance:

- Removes Reviewer Bias: Standardizes SOC quality assessment across divergent sectors (power, banking, etc)
- Verifiable Resilience: Detects subtle execution gaps and negative space across national infrastructure.
- Zero Data Leakage: Operates strictly air-gapped, keeping sensitive sovereign incident data offline.

- Operational Gain:** Replaces 5% manual sampling with complete metadata audits.
- Process Integrity:** Automatically flags tickets closed without genuine investigation
- Visibility:** Proactively spots silent assets and monitoring blind spots.
- Air-Gapped Security:** 100% offline deployment with zero network exposure.

Supervisory Coverage



What This Enables

✓

Operational Efficiency
Far less manual sampling needed to reach the same assurance quality.

✓

Consistency
No reviewer-to-reviewer variability — every entity judged by the same standard.

✓

National Resilience
Stronger oversight nationwide — consistent visibility across every Critical Sector Entity.

✓

Built to Scale
Grows with entity count and data volume — no added supervisory hours required.

NCIIPC Supervisors

- Scalable coverage across growing Critical Sector Entities
- Manual-review-level assurance, without the manual effort
- Consistent judgment — removes reviewer-to-reviewer variability
- Evidence-backed flags, not a black-box score

National Impact

- Stronger cyber resilience oversight nationwide
- Scales with entity count and data volume, no added hours
- Fewer manual sampling hours, more time on judgment calls
- Builds a track record of SOC process quality over time

@SAT-SA

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Unsupervised Learning Framework for Cyber Threat Detection

MDPI Applied Sciences, 2026

Isolation Forest, LOF, One-Class SVM

ML4SOC: A Comprehensive Review on ML for Security Operations

ACM Computing Surveys

Broad survey of ML in SOC contexts

ML for SOC Optimization & Critical Infrastructure Adoption

Springer

ML adoption & assessment in SOC

All three focus on detection speed —
how fast and accurately a SOC catches a threat, not how well it operates

The Gap SAT-SA Fills

No published work audits a SOC's own process quality — execution gaps and negative space —
rather than the threats it detects.

**This meta-supervisory framing is SAT-SA's original contribution —
built on the established, explainable ML techniques cited above.**